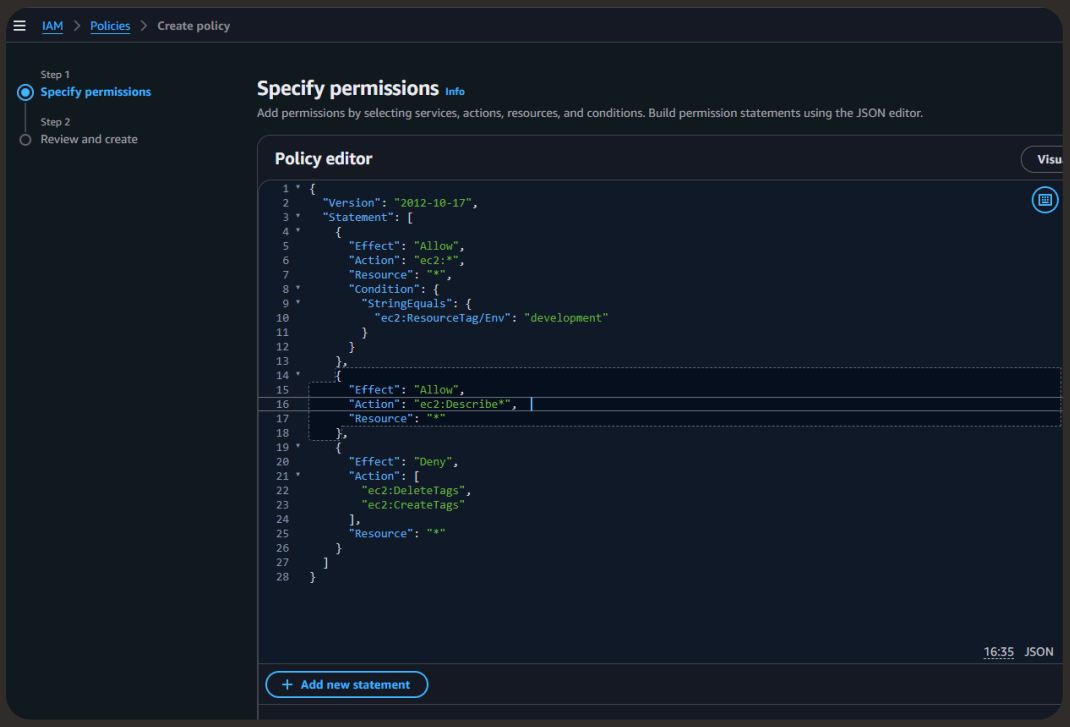
CLOUD SECURITY WITH AWS IAM

By Reynaldo Eusebio | September 2026



Introducing Today's Project!

Project overview

In this project, I will demonstrate how to use AWS IAM to control access to AWS resources. I'm doing this project to learn how to create IAM policies and manage permissions for users and groups.

Tools and concepts

Services I used were launching EC2 instances, using EC2 tags, creating IAM policies, creating IAM users and groups, and testing IAM access.

Key concepts I learnt include how to use IAM policies to control access to AWS resources, how tags can be used to manage permissions, and how to give users access to a development environment while preventing access to production.

Project reflection

This project took me approximately an hour.
The most challenging part was creating the IAM policy and understanding how the permissions worked.
It was most rewarding to see the policy work correctly by allowing access to the development instance while denying access to the production instance.

Tags

What I did in this step

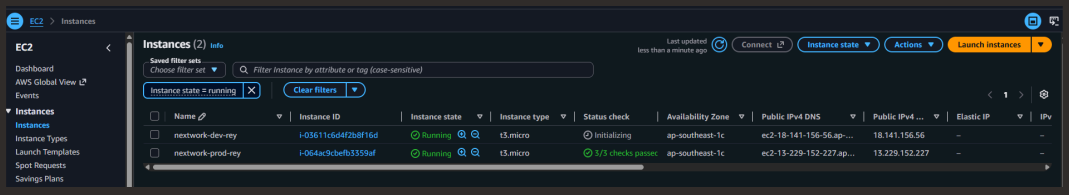
In this step, I will launch two Amazon EC2 instances because NextWork needs to increase its computing power to handle increased website traffic.

Understanding tags

Tags are used to easily find and identify EC2 instances.

My tag configuration

The tag I've used on my EC2 instances is called Env. The values I've assigned to my instances are production and development



IAM Policies

What I did in this step

In this step, I will create an IAM policy because it will give the intern the appropriate permissions to access and manage the development environment while preventing access to the production environment.

Understanding IAM policies

IAM policies are used to define permissions for who can access AWS resources and what they can do.

The policy I set up

For this project, I've set up a policy using JSON.

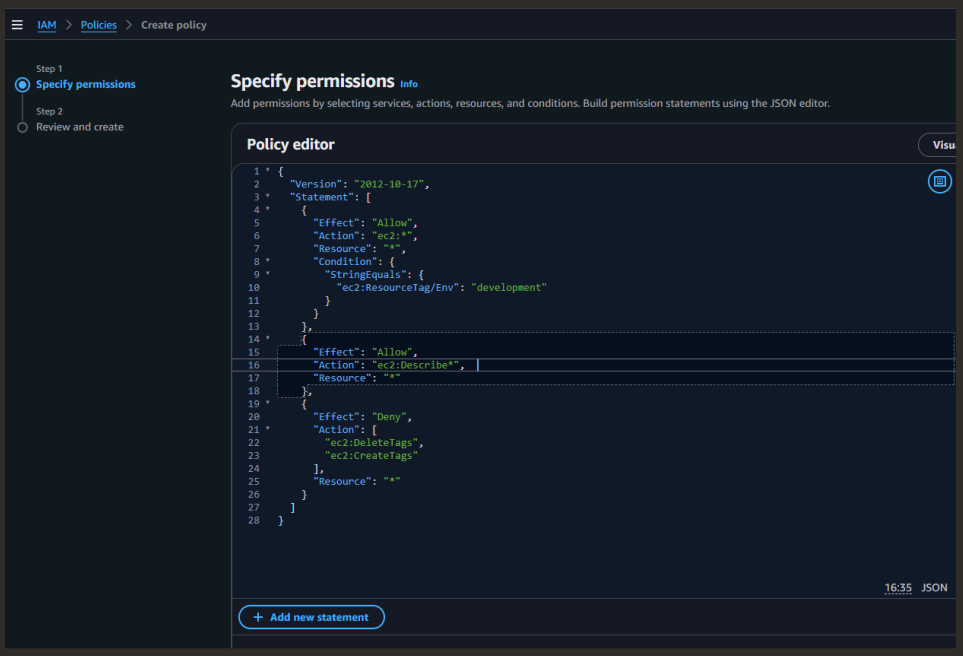
Policy effect

I've created a policy that allows the user to perform EC2 actions on resources tagged with Env=development, allows them to view EC2 resources, and denies them from creating or deleting tags.

Understanding Effect, Action, and Resource

The Effect, Action, and Resource attributes of a JSON policy mean that Effect specifies whether the action is allowed or denied, Action specifies what the user can do, and Resource specifies which AWS resources the action applies to.

My JSON Policy



Account Alias

What I did in this step

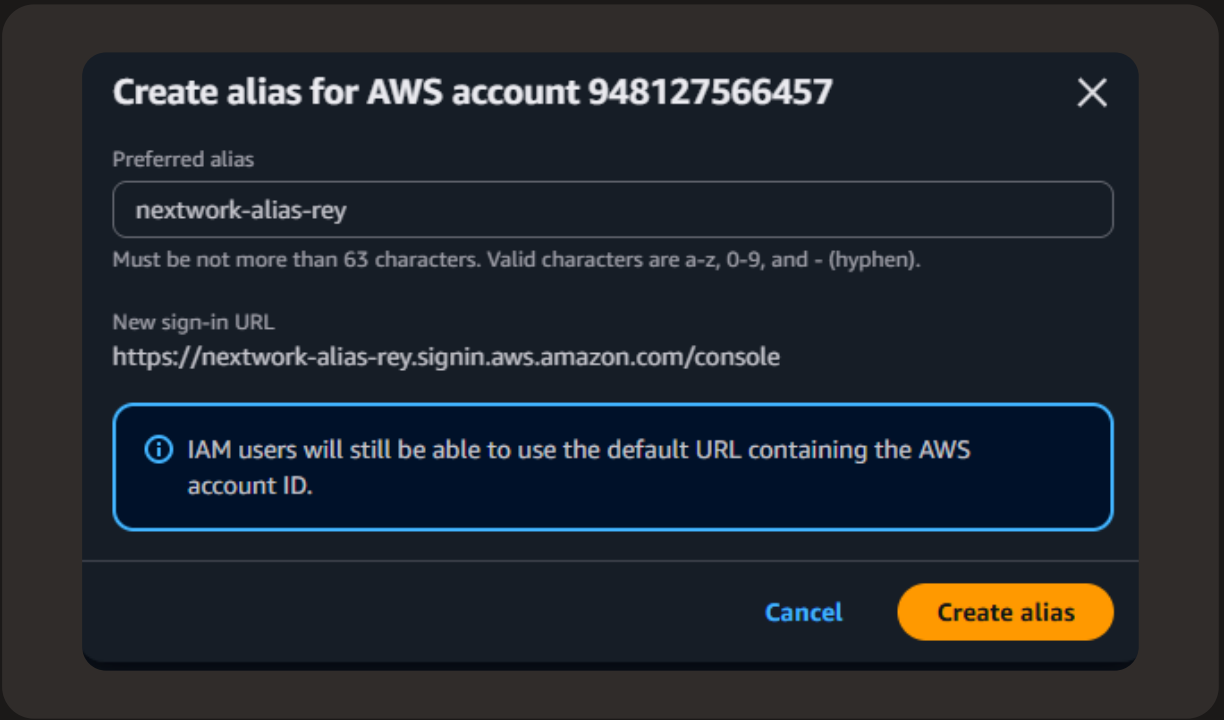
In this step, I will create an AWS account alias because it simplifies the user login process.

Understanding account aliases

An account alias is a human-readable ID for an AWS account.

Setting up my account alias

Creating an account alias took me a few seconds. Now, my new AWS console sign-in URL is https://nextwork-alias-rey.signin.aws.amazon.com/console.



IAM Users and User Groups

What I did in this step

In this step, I will create an IAM user and user group to give the intern access to the development environment but not the production environment.

Understanding user groups

IAM user groups are collections of IAM users.

Attaching policies to user groups

I attached the policy I created to this user group, which means that any user added to the group will be granted the permissions defined in the policy.

Understanding IAM users

IAM users are identities created for individual people or applications that need access to AWS resources.

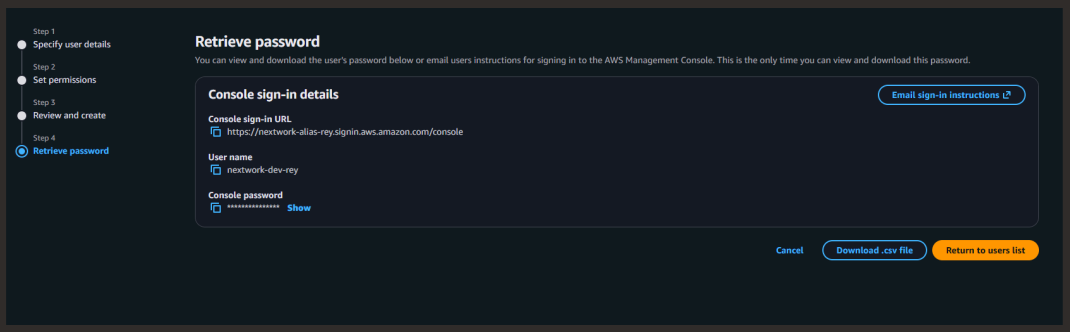
Logging in as an IAM User

Sharing sign-in details

The first way is by giving the user the sign-in details directly. The second way is by downloading the sign-in details as a .csv file.

Observations from the IAM user dashboard

Once I logged in as my IAM user, I noticed some “Access Denied” messages. This was because of the permissions defined in the IAM policy, which restricted what the user could access and do.



Testing IAM Policies

What I did in this step

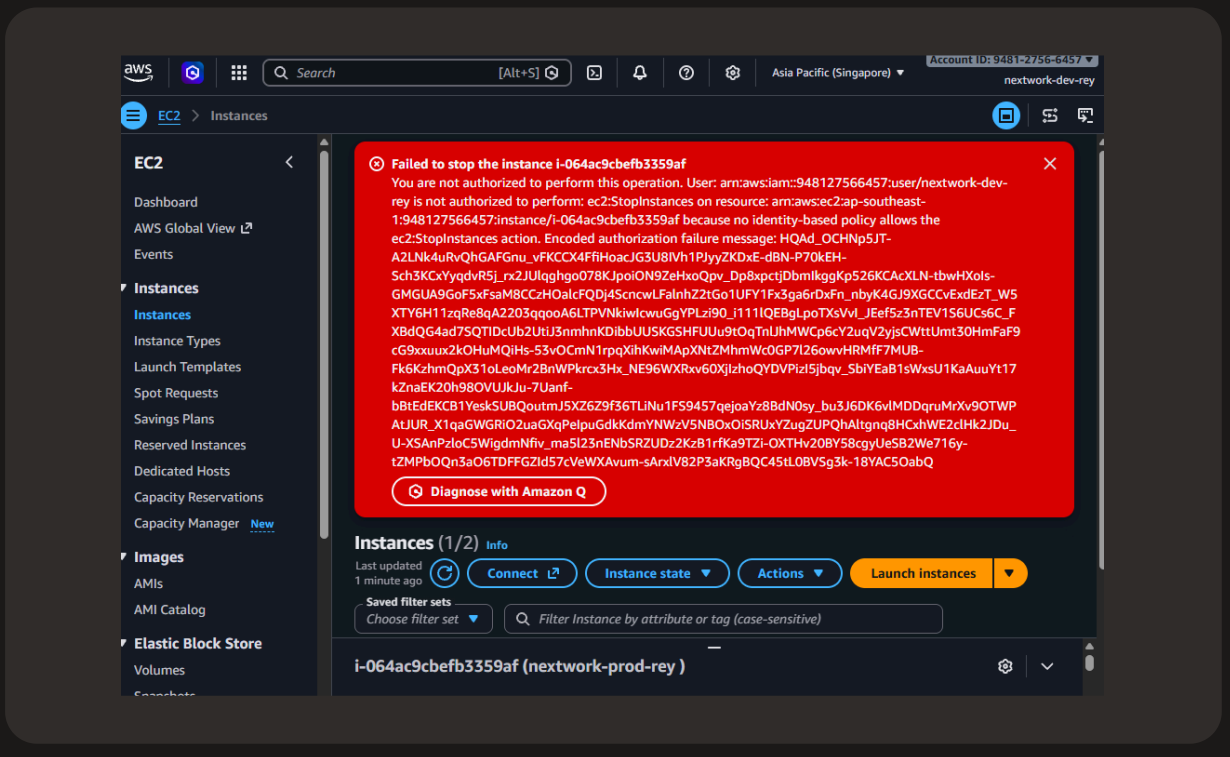
In this step, I will test the intern's access because I want to check if the IAM policy is working correctly and make sure the intern can access the development environment but not the production environment.

Testing policy actions

I tested my JSON IAM policy by trying to stop the production instance, but it didn't stop. When I tried to stop the development instance, it stopped successfully.

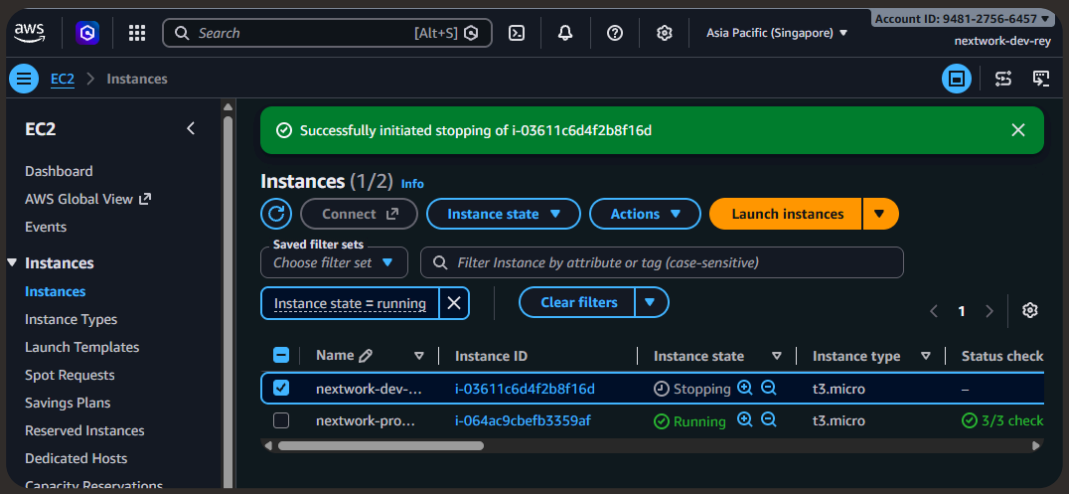
Stopping the production instance

When I tried to stop the production instance...it show error This was because...of the policy.



Stopping the development instance

Next, when I tried to stop the development instance, it stopped successfully. This was because the policy allows actions on EC2 instances tagged with Env=development.


aws
Search
[Alt+S]
Asia Pacific (Singapore)
Account ID: 9481-2756-6457
nextwork-dev-rey
EC2
Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Successfully initiated stopping of i-03611c6d4f2b8f16d
Instances (1/2) Info
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters
1
Name
Instance ID
Instance state
Instance type
Status check
nextwork-dev-...
i-03611c6d4f2b8f16d
Stopping
t3.micro
–
nextwork-pro...
i-064ac9cbefb3359af
Running
t3.micro
3/3 check

IAM Policy Simulator

To extend my project, I'm going to use the IAM Policy Simulator. I'm doing this because it is a best practice to test and verify that my IAM policy works as intended.

Understanding the IAM Policy Simulator

The IAM Policy Simulator is a tool used to simulate and test IAM policies. It's useful for checking whether a policy allows or denies specific actions.

How I used the simulator

I set up a simulation for the EC2 StopInstances action. The initial result was “Denied.” I had to adjust the simulation because the policy only allows stopping instances with the Env=development tag. I specified development in the EC2 instance tag field and ran the simulation again. The result was “Allowed.”

